24PY901

Quantum Principles, Structures and Computing

Module 1 – Unit 1

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Assignment 1: Nature of Light

Interaction with matter, Coherence, Polarization

References: Avadhanulu, Chapters 25. Atomic Physics, 3. Light, 10. Polarization

Conceptual questions

1. Explain light–matter interaction with reference to absorption, spontaneous emission, and stimulated emission.
2. Describe atomic spectra. Differentiate between emission spectra and absorption spectra with suitable examples.
3. State Bohr's quantization principle. Using Bohr's model, explain the origin of spectral lines in hydrogen atom.
4. Define coherence. Explain temporal coherence and spatial coherence. How are coherence time and coherence length related?
5. What is polarization of light? Explain linear, circular, and elliptical polarization.

Multiple choice questions

1. The energy of a photon is given by
   (a) $E = mc^2$

(b) $E = \frac{1}{2}mv^2$
(c) $E = h\nu$
(d) $E = qV$

2. Stimulated emission is important because it
   (a) produces incoherent radiation
   (b) emits photons randomly
   (c) produces photons identical in phase and direction
   (d) absorbs photons

3. Atomic spectra are discrete because
   (a) atoms are unstable
   (b) energy levels are quantized
   (c) atoms emit continuously
   (d) electrons move randomly

4. Bohr's quantization condition applies to
   (a) linear momentum
   (b) angular momentum
   (c) kinetic energy
   (d) potential energy

5. Coherence is related to
   (a) amplitude only
   (b) phase correlation
   (c) intensity only
   (d) frequency only

6. Spatial coherence refers to correlation between
   (a) Fields at different times
   (b) Fields at different spatial points
   (c) Frequencies
   (d) Polarizations

7. Temporal coherence is associated with
   (a) source size
   (b) bandwidth

(c) polarization
(d) intensity

8. Polarization of light proves that light is

    (a) longitudinal
    (b) scalar
    (c) transverse
    (d) massive

9. A laser shows high coherence because it has

    (a) High intensity
    (b) Narrow spectral width
    (c) Large size
    (d) Multiple modes

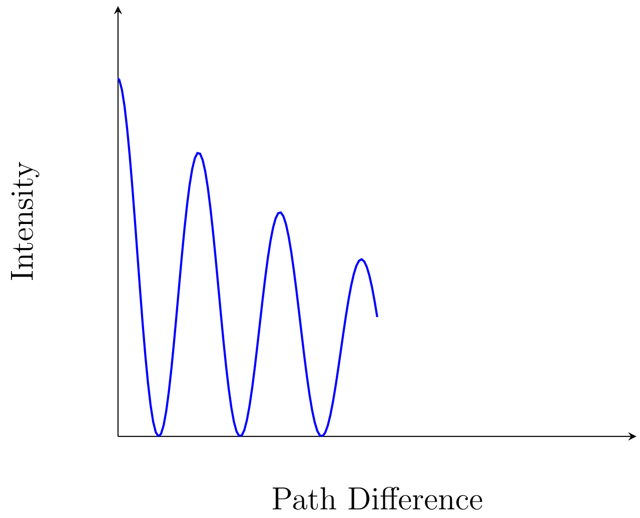
10. Consider two sources producing interference fringes. In experiment A, the fringe contrast decreases when slit separation is increased. In experiment B, fringe contrast decreases when path difference is increased.

    Experiment A is limited by ________ coherence and experiment B by ________ coherence.

    (a) Temporal, Spatial
    (b) Spatial, Temporal
    (c) Angular, Temporal
    (d) Spatial, Angular

Graph-based questions

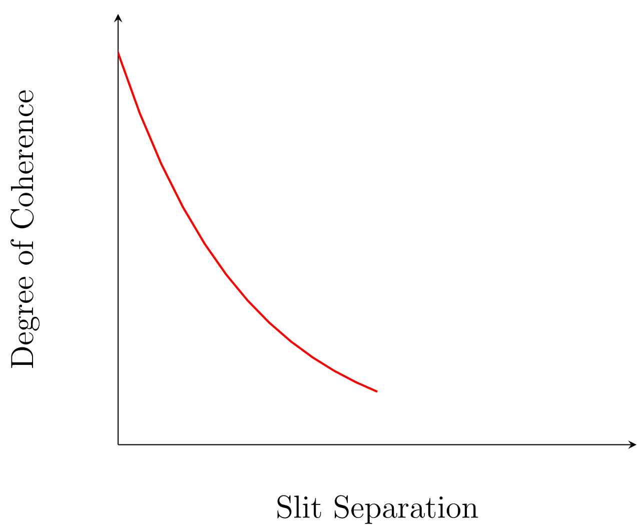
1. The following graph shows intensity variation with path difference.



The decrease in fringe visibility with path difference is due to

(a) Spatial coherence
(b) Temporal coherence
(c) Diffraction
(d) Polarization mismatch

2. Degree of coherence versus slit separation is shown below.

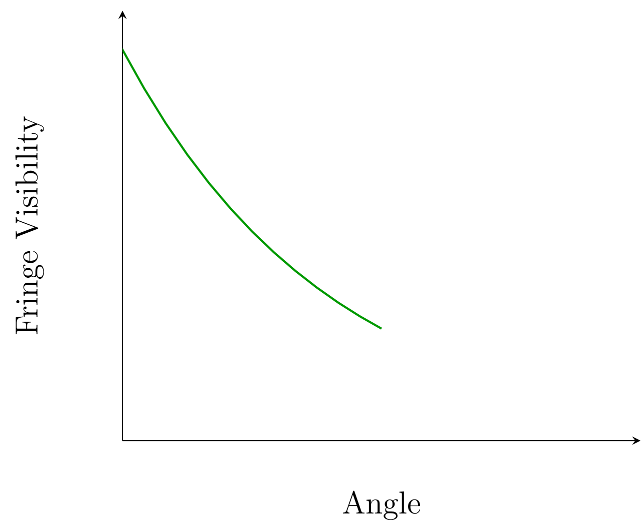


This graph represents limitation due to

(a) Temporal coherence
(b) Spatial coherence
(c) Angular dispersion

(d) Detector resolution

3. Fringe visibility variation with angle is shown.



This behavior is associated with

(a) Temporal coherence
(b) Spatial coherence
(c) Angular coherence
(d) Spatio-temporal coherence

Numerical questions

1. A source has spectral width $\Delta\nu = 5 \times 10^{11}$ Hz. The coherence time is

(a) 2×10^{-12} s
(b) 5×10^{-11} s
(c) 2×10^{-10} s
(d) 5×10^{-9} s

2. If coherence time is 10^{-9} s, coherence length is

(a) 0.03 m
(b) 0.3 m
(c) 3 m
(d) 30 m